

Troubleshooting Network Security Issues — Guided Notes

ANSWER KEY

Unit 5.3.2 | CompTIA Network+ N10-009 Obj. 5.3 | N10-008 5.3

For instructor use / distribute after students complete the notes.

Question 1

ARP poisoning and a duplicate IP address both produce intermittent connectivity and changing ARP cache entries. The key difference in ``arp -a`` output: duplicate IP shows two MACs _____ for the same IP, while ARP poisoning shows the _____ IP being claimed by an attacker's MAC.

Answer: Competing / alternating; gateway's (default gateway).

Question 2

A rogue DHCP server is detected when a client's ``ipconfig /all`` shows a _____ field listing an IP that is not your legitimate DHCP server. On the switch, the command _____ can identify which port the rogue device's MAC is connected to.

Answer: DHCP Server; `show mac address-table address`.

Question 3

To verify DNS hijacking, run ``nslookup`` against your configured DNS server, then run the same query against _____ (a known-good external resolver). If the results _____, your local DNS server is returning malicious or incorrect records.

Answer: 8.8.8.8 (or 1.1.1.1 or any trusted public resolver); differ.

Question 4

In firewall logs, a vertical port scan appears as one _____ IP hitting _____ destination ports on the same host in rapid succession. A horizontal scan appears as one source IP hitting the _____ port across many different destination IPs.

Answer: Source; sequential (many different); same.

Question 5

A compromised host may show three behavioral indicators in network data: (1) unexpectedly _____ outbound traffic volume, (2) connections to _____ or rotating external IP addresses not associated with known services, and (3) DNS queries for randomly generated _____ names (DGA).

Answer: High; unusual / unfamiliar; domain.

Question 6

A certificate warning showing 'certificate issued by untrusted authority' on a site that normally loads cleanly may indicate a _____ attack where an attacker is presenting a forged certificate. This warning should trigger immediate _____ rather than clicking through.

Answer: Man-in-the-middle (MITM) / SSL interception; investigation (escalation to security team).

Question 7

When a suspected compromised host is identified, the first response action is to _____ the device from the network. You should NOT _____ the device, as this destroys volatile memory that may contain forensic evidence.

Answer: Isolate / disconnect (disable the switch port or disconnect Wi-Fi); reboot.

Question 8 — Real World Application

REAL WORLD: A teacher reports their browser keeps redirecting to unfamiliar sites when accessing the school portal. The network team confirms connectivity is fine and ping works. Help desk clears the browser cache twice with no improvement. Describe the three diagnostic commands that would identify the root cause within five minutes.

Answer: 1. `ipconfig /all` — check the DHCP Server field; if it shows an unexpected IP, a rogue DHCP server is the source. 2. `nslookup portal.school.edu` — check the responding DNS server; if it's not the school's server, DNS is being redirected. 3. `nslookup portal.school.edu 8.8.8.8` — compare results from a trusted resolver; if the IPs differ, the local DNS is returning hijacked records. Root cause: rogue DHCP server pushing a malicious DNS address.